

- 18** A ball falls towards the ground from point X. Point X is 2.4 m above the ground.

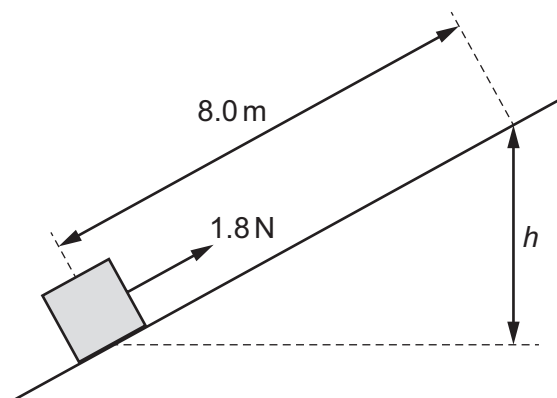
The ball rebounds from the ground and rises to point Y. Point Y is 1.8 m above the ground.

A single value of the change in height Δh is used to calculate the change in gravitational potential energy of the ball from X to Y.

What is the magnitude of Δh ?

- A** 0.6 m **B** 1.8 m **C** 2.4 m **D** 4.2 m

- 19** The diagram shows a block on a slope.



A constant force of 1.8 N in a direction parallel to the slope is exerted on the block. This causes the block to move along the slope at a constant speed.

The block moves a distance of 8.0 m along the slope and gains height h . The work done against the frictional force acting on the block is 4.8 J.

The weight of the block is 4.0 N.

What is the value of h ?

- A** 1.2 m **B** 2.4 m **C** 3.6 m **D** 4.8 m

- 20** The motor of a crane lifts a load of mass 600 kg. The load rises vertically at a constant speed of 12 m per minute.

What is the useful power output of the motor?

- A** 0.12 kW **B** 1.2 kW **C** 7.2 kW **D** 71 kW